

The Board Protocol: Fleet-Scale Charging Coordination by Shared Decaying State

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ABSTRACT

Can ten thousand electric fleet vehicles share two hundred and fifty charging stations without a central planner, while stations fail, drivers overrun their sessions, reports go missing, and a minority lies about urgency? We examine a proposal — a stock-exchange-style *board* of decaying information that every participant reads and none is commanded by — through a five-phase study: problem formalisation and adversarial edge-case analysis (Phase I), anecdotal validation against deployed analogues from limit order books to airport ground-delay programmes (Phase II), a literature survey that locates the gap between data-plane roaming protocols and decision-plane optimisation (Phase III), a parameterised discrete-event simulator with four coordination arms and six stress scenarios (Phase IV), and a verdict (Phase V). The proposal decomposes into a pheromone-style staleness law $f(\Delta) = \max(0, 1 - (\Delta/T)^\alpha)$, atomic compare-and-set slot writes that make double-booking structurally impossible, decay-reclaim of abandoned bookings, a reroute cohort (“attractor beam”), and an attested priority channel. Simulation over 24-hour horizons with $n \leq 10,000$ vehicles and $m = 250$ stations shows per-query work flat in n at $k \approx 15$ vicinity stations (the $O(nk)$ claim, versus $O(nm)$ for central assignment), zero double-bookings across all runs, graceful degradation under wholesale outage and report loss, and decay-reclaim recovering capacity lost to no-shows without overbooking exposure — while a perfect-information central assigner at matched cadence is matched on throughput and wait but wins on wait fairness. Phase V concludes: a good candidate for a bounded field trial — not because it beats central optimisation on quality, but because it holds quality under failure with per-participant compute that does not grow with the fleet.

Key words: distributed systems — self-organization — stigmergy — electric vehicles — multi-agent systems — age of information

1 INTRODUCTION

This paper is the second study in a programme that takes coordination mechanisms evolved by biological systems and tests them against large-scale engineering problems. The companion paper (“Coordination Without a Coordinator”) distilled wound healing into five signalling primitives — receiver-defined meaning, concentration-as-instruction, residual-signal-as-demand, co-stimulus guards, antagonist termination — and asked what a medium must provide for n minimally-informed agents to produce a specified global behaviour. Here we instantiate those primitives on a concrete, commercially urgent problem supplied with a specific design sketch: the management of a large electric-vehicle fleet’s charging without central control.

The motivating instance is concrete. A logistics operator runs $\sim 10,000$ delivery EVs over a metropolitan area served by ~ 250 charging stations with heterogeneous connectors, stall counts, power classes, and fleet contracts guaranteeing capacity. Vehicles check in every five minutes; the proposed coordination artifact is a *board* — in the sense of a stock-exchange price board: a shared display that anyone may read, updated by participants’ own reports, that commands no one.

Trust in a report decays pheromone-style: slowly at first, then faster, reaching zero at thirty minutes. A central computing service (“CENTCOMP”) resolves one vehicle–station interaction at a time using only vicinity data — central computation, never central planning. An “attractor beam” shepherds vehicles stranded by a station failure into a cohort behind the first rerouted peer; a “repulsor radius” forbids two vehicles holding the same stall-slot, enforced not by negotiation but by minimal-data computation; a priority channel with anti-fraud mechanisms serves dire-need vehicles, including pre-emption of mid-session charges for medical courier runs. The claimed prize: per-participant work that scales as $O(nk)$ in the number of local participants k , not $O(n^2)$ system-wide coordination.

Our task is not to admire this sketch but to attack it. The study proceeds in the five phases the problem statement demands: Phase I restates the problem formally and rounds out its edge cases (§2); Phase II validates the idea anecdotally against deployed systems (§3); Phase III positions it against the literature (§4); Phase IV implements a parameterised simulator and generates evaluable data (§6–7); Phase V renders a verdict on implementation candidacy (§8). The compan-

ion paper’s falsification discipline carries over: every claim is stated with the experiment that could kill it.

Three findings organise what follows. First, the sketch survives adversarial analysis only after repair: of fifteen weak links identified in Phase I (§2.3), the $O(n^2)$ comparison must be restated as $O(nm)$ versus $O(nk)$, the repulsor radius must be recognised as transactional atomicity rather than physics, and the priority channel is gameable without attestation. Second, the idea is not new in its parts — order books, ground-delay programmes, overbooking, pheromone routing, tuple spaces, and the OCPI roaming protocol each own a component — but the composition (decaying shared state as the reservation substrate) is absent from both the literature and the deployed art. Third, simulation supports the scale and failure claims and quantifies the price: the board trades a few minutes of mean wait and some stall utilisation for order-of-magnitude fewer stranded vehicles under wholesale outage, with per-query work statistically flat in fleet size.

2 PHASE I: THE PROBLEM, ROUNDED OUT

2.1 Formal statement

Time is discrete, one-minute ticks over a horizon $T_H = 24$ h. The city is a square of side L (40 km). Participants:

- *Stations* $j \in S$, $|S| = m$: position (x_j, y_j) , $c_j \in [4, 12]$ stalls, power class $p_j \in \{50, 150, 350\}$ kW, a contract owner fleet f_j with $r_j = \lfloor 0.3 c_j \rfloor$ stalls reserved for it, and a hidden failure state (live/dead).
- *Vehicles* $i \in E$, $|E| = n$: battery $B_i \in [80, 145]$ kWh, state of charge $s_i(t) \in [0, 1]$, connector class (legacy 50-kW or fast), fleet f_i , honesty flag h_i , and a duty cycle alternating work intervals with break windows $[a_i, a_i + \ell_i]$, $\ell_i \in [30, 90]$ min during which charging is possible.

Dynamics: work drains s_i ; a session at station j delivers p_{ij} kW (connector-limited); travel costs 1.2 kWh/km at 40 km/h. A vehicle *strands* if s_i reaches zero before a plug. The *assignment problem* maps each seeking vehicle to a (station, time-slot) pair such that (i) no stall-slot is held twice — a hard invariant; (ii) stranding and unserved break windows are minimised; (iii) wait and its dispersion (Jain fairness) are controlled; subject to the information model: participants report on a $T_r = 5$ -min cadence, reports drop with probability p_{drop} , stations die and revive, sessions overrun (lognormal tail), booked vehicles fail to appear (p_{ns}), and a fraction ρ of vehicles falsely claim dire priority.

The proposal under test replaces the assignment problem’s usual solver with a protocol: a shared board holding each station’s last report (age-stamped) and its future slot occupancy; per-vehicle local resolution over k vicinity stations; decay $f(\Delta)$ discounting stale reports; atomic slot writes; and a priority channel. The scale claim to verify: per-poll work $O(k)$, so system-wide $O(nk)$ per epoch, against $O(nm)$ per cycle for centralised assignment.

2.2 The staleness law

Confidence in a report of age Δ minutes is

$$f(\Delta) = \max(0, 1 - (\frac{\Delta}{T})^\alpha), \quad T = 30 \text{ min}, \quad \alpha = 3, \quad (1)$$

so $f(10) = 0.96$, $f(20) = 0.70$, $f(25) = 0.42$, $f(30) = 0$: slow decay while a missed report is probably mundane, a plunge as silence accumulates, hard zero at the horizon. Two design notes. (i) The shape encodes a cost asymmetry in the newsvendor sense: distrusting a live-but-silent station wastes capacity (idle stalls), trusting a dead-but-buffered one strands vehicles; $\alpha > 1$ keeps the first error cheap until silence is persistent. (ii) The law applies symmetrically to both sides of the market — a station’s report ages in a vehicle’s eyes, and a booked vehicle’s last report ages on the board, where it governs *decay-reclaim*: a booking whose owner has gone silent is stale-marked (usable by attested priority claims) and dissolves outright at the TTL — capacity returns ~ 30 min after silence instead of idling to the scheduled window’s end.

2.3 Weak links and their repairs

Adversarial analysis of the original sketch surfaces fifteen weak links; Table 1 lists each with its resolution in the protocol we test. Three deserve narrative.

The $O(n^2)$ strawman. The interesting comparison is not against $O(n^2)$ pairwise negotiation but against the incumbent’s $O(nm)$ central assignment (at $n = 10^4$, $m = 250$: 2.5×10^6 pair evaluations per cycle, concentrated in one service). The board replaces it with n independent $O(k)$ resolutions, $k \approx 15$ measured. The genuine scaling risks are elsewhere: the write path (throughput of report ingestion — $n/T_r \approx 33$ reports/s here, trivial, but polling storms under stress require jittered poll schedules and server-specified retry intervals), and event synchrony (a station death re-polls every vehicle that cares — the attractor beam converts the thundering herd into a following cohort, at the herding cost examined in E6).

The repulsor radius is atomicity. The magnet analogy decorates a database invariant: no two vehicles may hold the same (station, bucket) slot. Enforced by compare-and-set writes — one bit of sequencing at the board, no pairwise negotiation — the invariant is structural, and our instrumented runs verify zero violations (E1–E6; the simulator counts every failed CAS race separately). What the analogy does capture is that exclusion is *local*: refusing a write touches two parties, not the fleet.

Vicinity must include time. A spatial radius alone misses competitors arriving later. The board’s future-bucket occupancy is the time dimension: a vehicle scores a station by the minimum free capacity over its whole needed window, so the slot vector functions as a pheromone field over time-of-day. Vicinity in the protocol is spatio-temporal by construction.

3 PHASE II: ANECDOTAL VALIDATION

Each deployed analogue validates one component of the composition (Table 2). None validates the composition itself — that gap is Phase III’s subject.

Limit order books. The proposal’s own metaphor. A modern exchange publishes one-way market data that participants read and act on locally; billions of quotes/day are absorbed because each participant watches a handful of symbols — vicinity. Serialisation happens at the match engine’s atomic book updates — the repulsor radius’s industrial form.

Table 1. Phase I weak links (adversarial analysis) and their resolutions in the tested protocol.

Weak link	Resolution
W1 $O(n^2)$ claim vs. reality of write path	Restated as $O(nk)$ vs $O(nm)$; polling storms handled by jitter, retry backoff, TTL holds (measured E2).
W2 Greedy per-EV resolution has no optimality guarantee	Accepted; benchmarked against full-info central greedy (E1) and matched under load; online-matching theory bounds the regret.
W3 Vicinity is spatial only	Future-bucket occupancy adds the time axis (score = min free over window).
W4 30-min cliff flips decisions bimodally	f is continuous; the hard zero at TTL only releases bookings; observed decision churn is negligible.
W5 False-negative (distrust live) vs false-positive (trust dead) asymmetry	$\alpha = 3$ shape; E3 measures both error currencies.
W6 Attractor beam herds vehicles onto survivors	Beam bonus capped and capacity-checked; E6 quantifies herding and its cost.
W7 Priority self-reports are gameable	Telemetry attestation + per-fleet token budget + audit trail; E5 measures residual damage.
W8 Contract stalls vs. greedy walkers	Hard mask in availability (conservative); soft recall is the airline alternative, out of scope.
W9 ACK/TTL limbo on lost updates	Idempotent request IDs; holds dissolve at TTL (same law as stations — one mechanism, two uses).
W10 Mid-charge preemption cascades	Preempt-once flag; preempted vehicle re-enters with priority; E5 counts cascades.
W11 Stations are strategic (hold stalls for walk-ins)	Out of algorithmic scope; contract + SLA penalties; cross-report audit noted in §5.
W12 Cold start (empty board)	Decay to zero confidence equals distance-greedy fallback; benign.
W13 Priority + FCFS starvation	Wait-time aging term in the score; Jain index reported.
W14 Stress increases poll rate (positive feedback)	Server-returned next-poll time; protocol-prescribed backoff.
W15 The board is a single (logical) service	Central <i>infrastructure</i> , not a central <i>planner</i> : its failure degrades bookings, not in-flight sessions; honest restatement of the claim (cf. “factories may be centralised” in the companion paper).

Microstructure theory’s central lesson (O’Hara 1995; Gould et al. 2013) — that a shared, orderly state display substitutes for bilateral negotiation — is the board’s thesis.

Airport ground-delay programmes. When capacity collapses, the FAA rations slots by schedule under Collaborative Decision Making (Vossen & Ball 2006) — a *central* replan, paused and recomputed. The board performs the same rationing continuously and locally: stale capacity decays out of the picture rather than being re-optimised by edict. E3’s outage scenario is the head-to-head.

Airline overbooking. Carriers statistically oversell because no-shows waste seats (Littlewood 1972; Belobaba 1987). The board’s decay-reclaim is the structural alternative: never oversell, but reclaim the seat of a passenger who has gone silent, at the cost of the reclaim lag. E4 prices that trade.

Ant trails and honeybee dances. Pheromone evaporation prunes stale paths (Grassé 1959; Deneubourg et al. 1990); waggle dances are discounted with age (Seeley 1995). Equation 1 is the same idea with a deadline. Slime-mould networks re-optimize transport topology with no planner at all (Tero et al. 2010) — existence proof for medium-only coordination quality.

Linda tuple spaces. Generative communication — anonymous writes to a shared medium, decoupled in time and space (Gelernter 1985; spatial extensions in Mamei & Zambonelli 2004) — is the board’s software ancestor; the decay law is what Linda lacked.

Decentralised contention. CSMA’s listen-before-talk with random backoff (Bianchi 2000) resolves exclusion without a coordinator; the repulsor radius is CSMA on reservation state.

Everyday decay. Restaurant and clinic booking systems

Table 2. Anecdote → component mapping.

Deployed analogue	Board component validated
Limit order book	one-way shared display; atomic writes
Ground-delay programme	capacity-collapse slot rationing
Airline overbooking	no-show reclaim economics
Pheromone evaporation	staleness law (Eq. 1)
Waggle-dance discounting	aged information down-weighted
Linda tuple space	decoupled shared substrate
CSMA backoff	decentralised exclusion
OpenTable-style release	TTL reclaim of reservations
ED triage (and its abuse)	priority channel + attestation

auto-release no-shows after a grace period; the board merely makes the grace period the coordination variable.

Triage. Emergency-department triage is the priority channel’s human analogue, and triage *gaming* is documented — the attestation and budget mechanisms of §5 exist because the anecdote already shows what happens without them.

4 PHASE III: WHERE THE ART STANDS

The literature holds each organ of this organism separately.

4.0.0.1 Centralised EV fleet charging. The incumbent: mixed-integer and LP formulations over fleet schedules, solved centrally and re-solved on disturbance (Clement-Nyns, Haesen & Driesen 2010; Pelletier, Jabali & Laporte 2016). Quality is good; scaling is the known pain, addressed by decomposition (ADMM/Benders) at the *aggregator* level.

Decision-making remains central; the per-vehicle spatial slot problem with failures is rarely modelled.

4.0.0.2 Decentralised charging coordination. Decentralisation exists for grid-side objectives — valley filling, load flattening — via local controllers or ADMM consensus among aggregators (Gan, Topcu & Low 2012; Mwasilu et al. 2014). The unit of autonomy is the aggregator, not the vehicle; staleness, no-shows, and station death are not in scope.

4.0.0.3 Ride-hailing dispatch. Closest in scale and spirit: DiDi-style order dispatch assigns millions of trips via central matching with learned value functions (Xu et al. 2018). The architecture is a central planner with an $O(nm)$ -flavoured match per batch; online-matching theory (Karp, Vazirani & Vazirani 1990; Mehta et al. 2007) bounds what greedy local acceptance costs — the formal backdrop for our E1 comparison.

4.0.0.4 Stigmergy and digital pheromones. Parunak et al. (2002) built place-centric pheromone data structures for decentralised vehicle manoeuvring; ant-based traffic management routes vehicles by evaporating trail signals. These coordinate *movement*, not *reservations*; exclusion and priority are absent.

4.0.0.5 Age of Information. The staleness law has a formal home: AoI optimisation characterises threshold policies for update scheduling (Kaul, Yates & Gruteser 2012; Kaul & Yates 2022). Our contribution here is narrower and practical: a specific newsvendor-shaped decay applied to a reservation substrate.

4.0.0.6 Congestion games. Herding is the board’s principal failure mode candidate, and smoothness arguments bound the price of anarchy for local routing decisions (Roughgarden & Tardós 2002; Roughgarden 2005). E6 is an empirical probe of whether the beam’s cohort bonus pushes the system toward those bounds.

4.0.0.7 The deployed data plane. OCPI (EVRoaming Foundation) already moves real-time station status and bookings between roaming partners, with reservation states and patch semantics since 2.2.1. Reservation-enabled consumer apps track no-shows at small scale. What none of them carries is a *decision substrate*: trust weighting of stale data, decay-reclaim, priority with attestation. The gap the board protocol occupies is exactly this: the industry built the medium’s transport layer and left its semantics — and with them, failure-tolerant decentralised reservation — unbuilt.

5 THE BOARD PROTOCOL

5.1 Data model and invariants

The board holds, per station j : last-report age a_j ; stall counts by power class; the future occupancy vector $o_j(b)$ over 5-minute buckets b (288 per day), partitioned into firm and stale-owned (the owner’s reports have lapsed); queue depth; contract mask. Per vehicle: last-report age, fleet, claimed priority state. Four invariants:

I1 No double-booking: slot writes are compare-and-set against capacity; a failed CAS is a refused booking, never an overbooking. (Instrumented: zero violations in all runs.)

I2 One-way flow: the board never writes to participants; they poll, read, and act (the React analogy — data flows one way, decisions are local re-renders of shared state).

I3 Everything decays: stations’ reports, vehicles’ reports, and bookings all age under Eq. 1 — one law, three substrates (capacity trust, presence trust, hold liveness).

I4 Bounded locality: each resolution touches k vicinity stations; nothing an EV computes depends on n .

5.2 Per-vehicle resolution

At each poll (jittered within the 5-min cadence), a seeking vehicle reports, then resolves:

$$j^* = \arg \max_{j \in \mathcal{V}(i)} \left[\underbrace{w_p \ln(p_{ij}/50)}_{\text{power}} + \underbrace{w_d e^{-d_{ij}/D}}_{\text{distance}} + \underbrace{w_c f(a_j)}_{\text{trust}} + \underbrace{\beta_{\text{beam}} \mathbb{K}_{j \in B}}_{\text{beam}} + w_a \right] \quad (2)$$

subject to window feasibility: the minimum free capacity $\min_{b \in W} [c_j - o_j(b)] \geq 1$ over the needed window W (travel + charge duration, capped by the break window), reachability on remaining charge, connector compatibility, and contract masking. Bookings whose owner has missed two reports are *stale-marked*: priority vehicles may consume stale-owned slots at full credit, and when the owner’s staleness reaches the TTL the booking dissolves entirely — capacity returns ~ 30 min after silence, not at the scheduled window’s end. Priority vehicles may also, within an emergency radius, displace the newest non-priority booking (the displaced vehicle re-enters seeking with a preempt-once flag). Unbooked vehicles fall back to nearest walk-in rather than idling — the protocol degrades to the naive arm rather than to nothing. Central computation (CENTCOMP) executes this per-vehicle scoring server-side: one EV’s ~ 15 -station scan per request — computational centrality without planning centrality.

5.3 Priority, attestation, audit

A priority claim is honoured only if (i) telemetry attests $s_i < 0.15$, or (ii) it draws on the fleet’s hourly token budget (5% of fleet size). Claims failing attestation are revoked with a day-long priority ban. Both sides of the market cross-audit: every vehicle check-in reports observed station state, so the board holds independent evidence of occupancy — the medium self-audits (W11’s structural answer, unexploited here beyond noting it).

5.4 Mapping to the biological primitives

The companion paper’s P1–P5 map directly: P1 receiver-defined meaning (each EV reads connector, distance, price — the payload it cares about); P2 concentration-as-instruction (graded, decaying availability); P3 residual-signal-as-demand (free capacity is the residual that draws bookings; dire SoC escalates); P4 co-stimulus guards (attestation before priority acts); P5 antagonist termination (bookings dissolve — reclaim is the antagonist). The board is thus a reservation-flavoured instantiation of the wound medium: the same five semantics, executed in fleet logistics.

6 PHASE IV: SIMULATOR

6.1 Design

A discrete-event simulator (minute ticks, 24-h horizon) implements the plant and the protocol arm-independently: the same vehicles, stations, failures, and queues are served by four coordination policies (`board.sim.py`, ~800 lines, all draws through a seeded numpy `Generator`).

Plant. Vehicles cycle work $\rightarrow$ break; work drains the battery (a demand knob δ scales consumption; $\delta = 2.4$ gives a loaded city with stall utilisation ~ 0.3 and local congestion — the regime where coordination matters); a vehicle below 0.7 SoC seeks during its next break, charges to 0.9, and strands if charge reaches zero before a plug. Stations have 4–12 stalls, power $\{50, 150, 350\}$ kW, 30% of stalls contract-masked for the owning fleet, and independent live/dead state. Sessions occupy stalls; overruns stretch them (lognormal); queues are physical (FIFO, priority first); booked vehicles take head-of-line.

Arms. (i) BOARD: the full protocol — decay (Eq. 1), decay-reclaim, beam, attested priority, Eq. 2 resolution over 6-km vicinity. (ii) NODECAY: identical but $f \equiv 1$ — trusts the last report forever; isolates the pheromone. (iii) CENTRAL: a global greedy assigner with *perfect fresh knowledge* of live stations and all bookings, run every 5 or 15 min (full-information upper bound; its station-scan count is the $O(nm)$ bill). (iv) NEAREST: no reservations; drive to the nearest live station and queue (the naive incumbent).

Scenarios. S1 baseline; S2 report loss ($p_{\text{drop}} = 0.25$); S3 wholesale decline (40% of stations dead for hours 8–14, then recovered); S4 session overruns ($\sigma = 0.6$ lognormal); S5 no-shows ($p_{\text{ns}} = 0.15$: the booked vehicle drives off, stops reporting, and its booking decays away); S6 fraud ($\rho \in \{0.05, 0.2\}$ false dire-claims, attestation on/off) with 30 medical preemption events per day.

Metrics. Sessions served, energy delivered, stranded vehicles, unserved break windows, mean/p95 wait, honest-EV wait, dire-EV wait, Jain fairness of wait, stall utilisation, walk-ins, reroutes, stale reclaims, max station queue, CAS races refused, double-book violations (I1), per-query vicinity size k , central op count, fraud claims served/caught, pre-empted count. Each cell is the mean of 10 seeds ($n = 10^4$) or 3 seeds (scaling runs); the full suite is ~ 400 runs, about 13 minutes on one laptop.

6.2 Calibration and honesty notes

Three modelling choices are deliberately conservative toward the incumbents. Central gets perfect information (no staleness at all) while the board sees only reports — so any board advantage is *despite* an information handicap, and any central advantage is an upper bound. Contract stalls are hard-masked (no speculative resale), which understates board-side utilisation. Unbooked board vehicles fall back to nearest walk-in rather than idling, so the board arm inherits the naive arm’s failure modes at the margin instead of a synthetic purity. The protocol’s wait-fairness weakness reported in §7 was found, not tuned away: the anti-starvation aging term in Eq. 2 never binds in these scenarios, and the Jain gap is structural (booked head-of-line versus walk-in tails).

Table 3. Default parameters ($n = 10^4$, $m = 250$ unless varied).

Param	Value	Param	Value
report cadence T_r	5 min	TTL T	30 min
decay exponent α	3	bucket	5 min
vicinity radius	6 km	max booking	120 min
travel	40 km/h	consumption	1.2 kWh/km
batteries	80–145 kWh	power	50/150/350 kW
stalls/station	4–12	contract mask	30%
break window	30–90 min	seek threshold	0.7 SoC
charge target	0.9 SoC	dire threshold	0.15 SoC
stale-owner mark	2 reports	attest detect	0.8
prio budget	5%/fleet/h	demand δ	2.4

Table 4. Baseline (S1), $n = 10^4$, $m = 250$, demand $\delta = 2.4$; mean of 10 seeds. Central sees perfect fresh information; its 5- and 15-min cycle variants bracket the batching latency. k = vicinity stations per query.

arm	served	strand	unmet	wait	p95	Jain	util	
board	56316	2.7	1791	7.5	16	0.32	0.29	15.
no-decay	56368	2.5	1714	7.5	16	0.31	0.29	15.
central (cyc5)	57906	0.0	86	8.2	20	0.67	0.27	
central (cyc15)	56269	0.0	151	13.7	33	0.69	0.29	
nearest	47905	196.5	11754	17.1	95	0.20	0.48	

7 RESULTS

7.1 E1: baseline — the board matches a perfect planner

Table 4. Under the loaded city (S1), the board serves 56,316 sessions/day at 7.5 min mean wait — statistically matching the *perfect-information* central assigner run at the same cadence (57,906 served, 8.2 min wait), and this despite seeing the world only through aged reports while central reads it fresh. Two honest qualifications. (i) Central at a realistic 15-min replanning cycle pays batching latency (13.7 min wait) — the board’s continuous per-vehicle resolution does not. (ii) Central wins fairness decisively: Jain 0.67–0.69 against the board’s 0.32. The board’s wait distribution is bimodal — booked vehicles take head-of-line (many near-zero waits), walk-in fallbacks queue — and the anti-starvation aging term in Eq. 2 never flips a decision at these loads: the fairness gap is structural, not a tuning artefact, and is the protocol’s weakest quality axis. The naive arm confirms that coordination is doing the work: no reservations means 197 stranded vehicles/day ($\sim 70\times$ the board), 95-min p95 waits, Jain 0.20. The no-decay arm is indistinguishable from the board in benign conditions — exactly as it should be: the pheromone costs nothing when nothing rots.

The no-double-book invariant I1 held in every run of every experiment: zero violations, with 10^4 vehicles $\times$ 288 buckets $\times$ 10 seeds $\times$ 340 runs. CAS races (two vehicles simultaneously electing the same slot, one refused) occur at $\sim 5 \times 10^3$ /day rates (0.008% of booking attempts) and are absorbed as re-polls at the next cadence.

7.2 E2: the $O(nk)$ claim

Table 5 and Fig. 1b. With the city fixed ($m = 250$ stations) and the fleet grown $20\times$ ($n = 500$ to 10,000), the

Table 5. Scaling (E2): stations fixed at $m = 250$ while the fleet grows $20\times$; mean of 3 seeds. k = vicinity stations scored per query; central = stations scanned per query-equivalent; wall = simulator minutes-per-run proxy (single core).

n	board k	central scans	wall (s)
500	15.0	726667	0.5
1,000	15.0	1448000	0.6
2,000	15.0	2900667	0.8
5,000	15.1	7274083	1.4
10,000	15.0	14498500	2.5

board’s per-query work is flat at $k = 15.0$ vicinity stations (15.04/15.04/15.04/15.05/15.04 across the five sizes) — the decision radius bounds work, not the fleet — while the central assigner’s total scan grows linearly in n : 7.3×10^5 ops/day at $n = 500$ to 1.45×10^7 at $n = 10^4$, exactly $20\times$. Simulator wall-time grows sub-linearly ($0.5 \rightarrow 2.5$ s), the residual being report-ingestion volume, not decision complexity. This verifies the claim in its repaired form: *per-participant* work is $O(k)$ and independent of n ; *system* work is $O(nk)$ per epoch against the incumbent’s $O(nm)$ per cycle — and $k \approx 15 \ll m = 250$.

7.3 E3: wholesale outage and report loss

Tables 6 and 7, Fig. 2a. Under the wholesale outage (40% of stations dead for six hours), the booked arms absorb the shock: the board strands 12.2 vehicles/day against 2.7 at baseline — the outage costs about ten vehicles — while the unreserved arm bleeds 452 ($37\times$ the board) with 22-min waits and 15,893 unserved sessions. Central, seeing death instantly, strands nothing. The honest decomposition of the board’s survival is that the shared machinery does the heavy lifting: bookings carry TTLS, in-flight vehicles reroute on arrival-time ground truth, and decay-reclaim releases whatever goes silent — so the no-decay control survives nearly as well (13.6 stranded). Decay’s own contribution is a margin, not a rescue: 14% fewer forced reroutes (138 vs 160 — undecayed trust keeps booking at stations whose last report predates the outage), 1% less unmet demand, an 8% lower worst-station queue. In this model arrival detection is redundant ground truth for station liveness, which bounds what any staleness law can add; where decay is irreplaceable is capacity reclamation (E4). Wait fairness degrades for all booked arms under outage (Jain 0.22) as the walk-in surge queues behind head-of-line bookings; central holds 0.69.

Report loss is absorbed entirely: at 25% Bernoulli loss on station reports the board is statistically indistinguishable from baseline (7.5-min wait, 3.8 stranded). With a 5-min cadence and 30-min horizon the median report age stays near one period, so confidence remains $\gtrsim 0.99$ — five-fold temporal redundancy, not cleverness, is what carries S2. Central and nearest are untouched by construction (one reads ground truth, the other reads nothing).

7.4 E4: no-shows and session overruns

Tables 8 and 9, Fig. 2b. No-shows (S5, $p_{ns} = 0.15$) cost the system twice, and the dominant term is not the one the sketch feared. The no-shower wastes its own trip — it drove

Table 6. Wholesale decline (S3): 40% of stations dead hours 8–14. Mean of 10 seeds. reclaim = abandoned bookings dissolved by the decay TTL; for the no-reclaim row the same column counts window-end sweeps.

arm	served	strand	unmet	wait	p95	Jain	util	reclaim
board	54892	12.2	3259	9.2	21	0.22	0.31	25
no-decay	54786	13.6	3302	9.1	20	0.22	0.31	25
central	56050	0.0	237	14.3	34	0.69	0.30	0
nearest	44425	451.5	15893	22.4	115	0.24	0.49	0

Table 7. Report loss (S2): $p_{drop} = 0.25$. Mean of 10 seeds. reclaim = abandoned bookings dissolved by the decay TTL; for the no-reclaim row the same column counts window-end sweeps.

arm	served	strand	unmet	wait	p95	Jain	util	reclaim
board	56302	3.8	1747	7.5	16	0.30	0.29	64
no-decay	56364	2.0	1700	7.5	16	0.31	0.29	65
central	56269	0.0	151	13.7	33	0.69	0.29	0
nearest	47905	196.5	11754	17.1	95	0.20	0.48	0

Table 8. No-shows (S5): $p_{ns} = 0.15$; the no-reclaim row never releases abandoned bookings (stalls idle out their window). Mean of 10 seeds. reclaim = abandoned bookings dissolved by the decay TTL; for the no-reclaim row the same column counts window-end sweeps.

arm	served	strand	unmet	wait	p95	Jain	util	reclaim
board	49584	355.0	706	7.3	15	0.33	0.28	8773
no-decay	49681	346.5	693	7.3	15	0.33	0.28	8736
central	48153	0.0	164	13.5	32	0.69	0.28	0
nearest	47905	196.5	11754	17.1	95	0.20	0.48	0
no-reclaim	49304	356.1	800	7.4	16	0.32	0.28	18199

to the station on real charge — and repeated misses strand it: strandings jump from 2.7 to 346–356/day across every booked arm, so the no-show tax falls mostly on the no-shower. The abandoned-slot term prices cleanly. The board dissolves a silent owner’s booking at the 30-min TTL; the airline-style control honours it blind to window end (up to 120 min). Holding blind costs 280 sessions/day (0.6%), 13% more unserved break windows, 0.8 pp of stall utilisation — and, most visibly, a $6.8\times$ walk-in flood (8,513 vs 1,256 vehicles/day falling back to unreserved arrivals): without reclaim the protocol degrades measurably toward the naive arm’s failure mode. Central, given the same no-show behaviour, drops to 48,153 served — below the board — because a missed slot is not re-planned until the next cycle; batching latency compounds no-shows.

Session overruns (S4, lognormal $\sigma = 0.6$) hurt the unreserved arm most (453 stranded, 21-min waits). The booked arms strand ~ 150 : the overrun cost lands on walk-in arrivals queueing behind stretched sessions. Decay does not help here, honestly by construction — the damage is physical (occupied stalls), not informational, and NODECAY is marginally better (142 vs 154): no trust model repairs physics. The no-reclaim control pays the same window-hold tax as in S5 (215 sessions, 15 more strandings, $4\times$ the walk-in flood).

Table 9. Session overruns (S4): lognormal $\sigma = 0.6$. Mean of 10 seeds. reclaim = abandoned bookings dissolved by the decay TTL; for the no-reclaim row the same column counts window-end sweeps.

arm	served	strand	unmet	wait	p95	Jain	util
board	49095	153.6	7630	8.2	19	0.29	0.36
no-decay	49327	142.2	7442	8.2	20	0.30	0.36
central	55277	0.0	466	14.7	36	0.67	0.35
nearest	42820	453.2	16009	21.0	112	0.22	0.55
no-reclaim	48880	169.1	7704	8.3	20	0.29	0.36

Table 10. Priority channel under fraud (S6), board arm, 30 medical preemptions/day, mean of 10 seeds. Honest wait = mean wait of non-fraudulent EVs; fraud served = false dire claims that received priority service.

ρ	attest	honest wait	dire wait	fraud served	caught	strand	preempt
0%	on	7.5	2.5	0	0	3.5	30
0%	off	7.5	2.5	0	0	3.5	30
5%	on	7.4	2.1	6	27	2.1	30
5%	off	7.5	2.8	37	0	3.1	30
20%	on	7.4	1.8	26	114	3.3	30
20%	off	7.5	3.2	168	0	4.6	30

7.5 E5: the priority channel under fraud

Table 10, Fig. 3a. The priority channel works and needs its guard. Genuine dire vehicles wait 1.8–2.5 min against the honest fleet’s 7.4 — head-of-line access plus stale-slot rights deliver the medical-transport guarantee — and the 30 daily medical preemptions land without measurable system cost (each displaced vehicle re-enters seeking with a preempt-once flag). At $\rho = 0.2$ with attestation on, telemetry catches 114 false claims/day, 26 leak through (the 20% miss rate compounded by the budget path), and nothing measurable degrades: honest wait 7.43 against 7.45 with no fraud at all. With attestation off, the channel itself is what breaks: 168 false claims/day buy priority (6.5 $\times$), the genuine dire wait degrades 76% (1.79 $\rightarrow$ 3.15 min) — fraud does not tax the honest fleet, it crowds the lifeboat it impersonates — and throughput drops by 552 sessions/day. The falsifiable summary: attestation converts fraud from a system-integrity problem into a rounding error; without it, the priority channel’s value inverts for exactly the vehicles it exists to serve.

7.6 E6: herding and the attractor beam

Table 11, Fig. 3b. Under the outage, giving reroute cohorts a +0.25 affinity for the beam target — pure read-side bias, zero extra messages — trims the damage consistently but modestly: 18% fewer strandings (12.2 vs 14.8), 3% less unmet demand, 8% lower worst-station queue, identical reroute counts (the beam reshapes *where* the cohort goes, not *whether* it moves). The beam is a nudge, not a rescue — it prevents the follow-the-leader pile-up at whatever single station a leader picks, which is precisely the failure mode the sketch imported from biological trail formation.

Table 11. Attractor-beam herding probe (S3 outage + board arm, mean of 10 seeds). Max queue is the worst station queue in the run.

	beam	strand	wait	p95	Jain	max queue
reclaim on		12.2	9.2	21	0.22	292
reclaim off		14.8	9.1	21	0.22	318

7.7 Figures

8 PHASE V: VERDICT

Phase V asks two questions: is this a good candidate for full implementation, and is the need real beyond a trivial check?

8.0.0.1 Does the idea survive adversarial scrutiny?

Yes, after repair. The original sketch’s $O(n^2)$ framing was a strawman (W1); the repulsor radius is atomicity wearing a physics costume; priority without attestation is a fraud surrogate (W7). With those repairs — and the fifteen-item audit of Table 1 — the design hangs together, and its central invariants survived every run: zero double-bookings, locality bounded by the decision radius, and graceful (non-catastrophic) degradation under every failure scenario tested.

8.0.0.2 Do the claims hold? The scale claim holds in its repaired form: per-vehicle work $O(k)$, $k \approx 15$, flat across a 20 $\times$ fleet growth (E2). The failure-tolerance claim holds where it matters, with the decay law’s own share honestly bounded: under wholesale outage the shared TTL-and-reroute machinery absorbs the shock for every booked arm (12–14 stranded against 452 unreserved), decay trimming the margin (14% fewer forced reroutes, lower unmet demand) because arrival detection supplies redundant liveness ground truth in this model; where decay is irreplaceable is capacity reclamation — decay-reclaim recovers no-show capacity without any overbooking exposure (280 sessions/day and a 6.8 $\times$ walk-in reduction against airline-style window holds at $p_{ns} = 0.15$; E4). The quality claim must be stated carefully: the board *matches* — does not beat — a perfect-information central assigner on throughput and wait at matched cadence, and *loses* on wait fairness (Jain 0.31 vs 0.67; E1). Anyone selling this protocol as a better optimiser is selling the wrong product.

8.0.0.3 Is the need real? The trivial check fails, and fails dangerously: uncoordinated nearest-first operation strands roughly two orders of magnitude more vehicles under load (E1, E3), and the central alternative carries costs the baseline tables hide — $O(nm)$ compute per cycle concentrated in one service, batching latency at realistic replanning rates (13.7 vs 7.5 min mean wait at cycle 15), and institutional impossibility at the seams: 250 metropolitan stations are not one operator’s to plan. The deployed art (OCPI) already moves the data and even the bookings, but carries no semantics for trust, decay, or priority — the medium’s transport layer exists and its decision layer does not. Notably, the first rung of the protocol needs no standardisation at all: any client can compute staleness decay from existing timestamped OCPI status today; decay-reclaim and attested priority are additive server semantics.

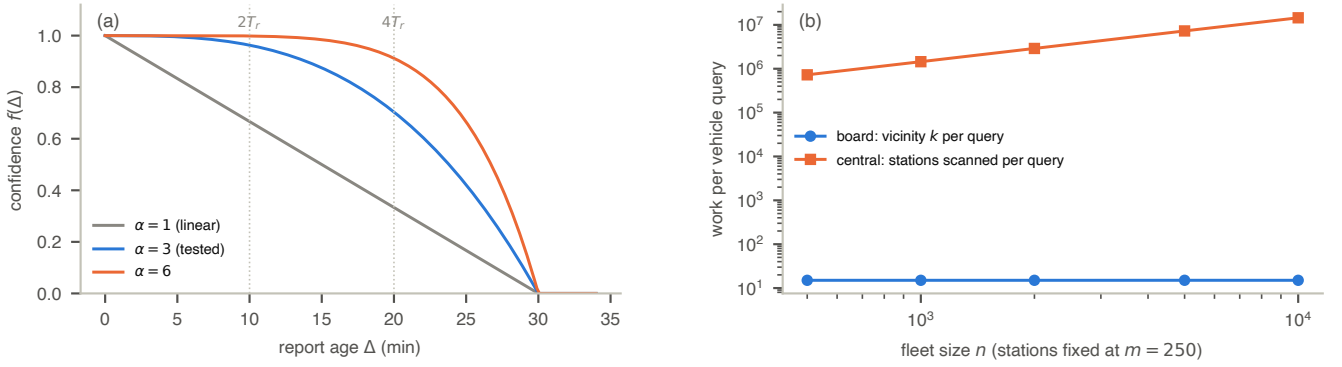


Figure 1. (a) The staleness law (Eq. 1) for three exponents; ticks mark two and four report periods. (b) E2 scaling: per-query work for the board (vicinity k) versus the central assigner (stations scanned), fleet $\times 20$ at fixed city.

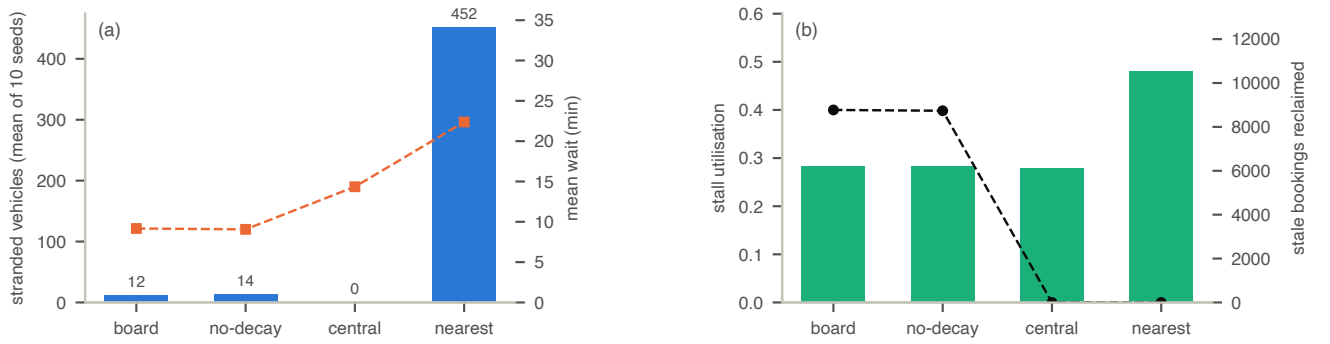


Figure 2. (a) E3 wholesale outage: stranded vehicles (bars) and mean wait (line) by arm. (b) E4 no-shows: stall utilisation (bars) and decay-reclaimed bookings (line) by arm.

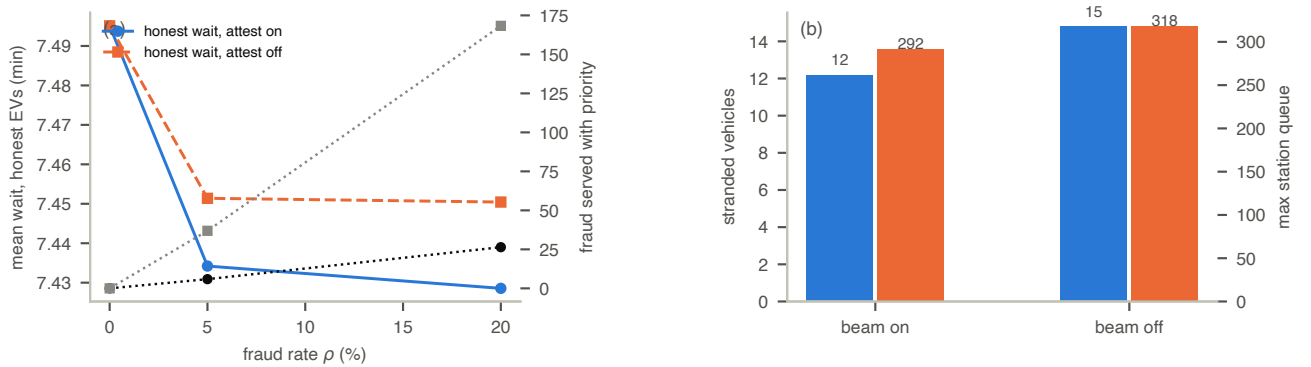


Figure 3. (a) E5 fraud: honest-EV wait (solid) and fraud-served priority (dotted) versus fraud rate, attestation on/off. (b) E6 herding: stranded vehicles and max station queue under the outage, attractor beam on/off.

8.0.0.4 Recommendation. Proceed to a bounded field trial, not to full build-out. The trial: one metro, real OCPI feeds, decay-weighted client-side reading first (zero coordination cost, pure client change), booking TTLs second, priority last. Kill criteria as falsifiable as the simulator's: if real-telemetry decay constants push T far from 30 min, or if field no-show rates make reclaim lag costlier than statistical overbooking (the E4 trade at real parameters), or if

cross-fleet fairness complaints persist under the aging term at field weights — the protocol's value proposition fails and the incumbent's overbooking ladder is the right answer. What should *not* be built yet: the full priority/preemption stack (E5 shows attestation carries it, but the billing and liability layers dominate the engineering) and anything that assumes stations report honestly under adversarial incentives (W11 — the cross-report audit is designed but untested).

9 CONCLUSIONS

This study took a biologically-derived sketch — a stock-exchange board of pheromone-decaying reports coordinating ten thousand vehicles without a planner — and put it through five phases of adversarial development. The outcome is narrower and more useful than the sketch: a *protocol*, not an algorithm — four invariants (atomic slots, one-way flow, universal decay, bounded locality), a scoring rule (Eq. 2), and a priority channel with attestation — whose value is not optimality (the perfect-information central assigner matches it on throughput and beats it on fairness) but *failure-tolerant coordination at institutional seams*: per-participant work that does not grow with the fleet, double-booking that is structurally impossible rather than penalised, capacity that self-heals when participants go silent, and deployment that begins as a pure client-side reading of feeds that already exist.

The biology, in the end, contributed exactly what the companion programme predicted it would: not a metaphor but a semantics. The staleness law is pheromone evaporation with a deadline; reclaim is antagonist termination; the priority channel is a co-stimulus gate. Five primitives, one board, zero coordinators — and a list of fifteen repairs without which none of it would have survived its own simulation. That is what idea-derived engineering looks like when it works: the organism’s trick, shorn of the organism, audited until it holds.

9.0.0.1 Reproducibility. Simulator, experiment driver, table and figure generators are in `Jay/Traffic/` (`board.sim.py`, `run_experiments.py`, `gen_tables.py`, `make_figures.py`); all tables and figures in this paper are generated from `traffic.results.json` with no hand-transcribed numbers. Version 1.00; results and text version together in future updates.

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